

SOP-116 — Cooling Water System Operation - Utilities Area

Plant Site A · Rev 1 · Issued 03/10/2024 · Approved by Plant Engineering

1. Purpose

This procedure defines normal operation of the Utilities Area cooling water system at Plant Site A, including pump duty rotation, basin level control and chemical dosing surveillance. Stable cooling water supply protects every heat exchanger and machine cooler on site, and most process temperature excursions trace back to cooling water disturbances that were preventable. The procedure also defines the surveillance readings that feed the reliability programme for the Utilities Area rotating equipment.

2. Scope

Applies to the cooling water circulation pumps P-102A and P-102B, the auxiliary pump P-103A, the cooling water return exchanger HX-202A, and the basin makeup control valve CV-303A. Cooling tower fan maintenance and chemical dosing pump calibration are covered by their own procedures.

3. References

SOP-101 Centrifugal Pump Startup and Shutdown Procedure.

SOP-110 Rotating Equipment Vibration Monitoring and Bearing Lubrication.

Cooling water chemical treatment control sheet issued by the treatment contractor.

4. Safety Precautions

Cooling tower basins and sumps are confined spaces when drained. Entry requires a confined space permit without exception.

Chemical dosing lines carry corrosive biocide and inhibitor concentrates. Wear chemical splash PPE when working near dosing points, and know the location of the safety shower.

Do not approach the fan deck while any fan is in automatic; fans may start without warning on temperature control.

5. Tools and Materials

Have available on rounds: the shift log or work tablet, the contractor chemical test kit with in-date reagents, chemical splash PPE for dosing point checks, a torch for basin screen inspection, and the utilities log sheet. A duty pump changeover additionally requires the SOP-101 checklist and radio contact with the panel operator.

Verify the test kit reagent expiry dates weekly. Residual readings taken with expired reagents have repeatedly caused false out-of-range reports and unnecessary contractor callouts.

6. Procedure

6.1 Operate the system with one main circulation pump on duty and one on standby autostart. The duty selection shall alternate between P-102A and P-102B on the first day of each month to equalise running hours across the pair.

6.2 Complete the changeover per SOP-101: start the incoming pump, confirm stable discharge pressure, then stop the outgoing pump. Never leave both pumps stopped with exchangers on load, however briefly.

6.3 Record the running hours of P-102A and P-102B at each monthly changeover and reconcile them quarterly. A growing gap in hours means the rotation is being skipped and one machine is quietly accumulating all of the wear.

6.4 Check supply header pressure and temperature each shift and record them in the shift log. A falling header pressure at constant pump speed indicates strainer or basin screen blockage and shall be investigated the same shift.

6.5 Maintain the basin level within the marked band using CV-303A on automatic makeup. If the valve is passing or hunting, place makeup on manual bypass control and raise a work request for the valve. Track basin makeup volume alongside the level record; rising makeup at a constant basin level indicates a circuit leak to be located and reported.

6.6 Verify the side-stream filter is in service and back-flushing on its timer. A side-stream filter left in bypass measurably accelerates fouling of HX-202A and of every user exchanger on the circuit.

6.7 Record the cooling water supply and return temperatures across HX-202A daily. A rising approach temperature at constant duty is the earliest indication of exchanger fouling and shall be reported to the reliability engineer for cleaning assessment against SOP-105.

6.8 Once per shift, listen at each running pump for changes in sound and check the seal area visually. Operators detect most developing pump problems days before the monthly instrument route does, and a verbal report is sufficient to trigger a check reading.

6.9 Confirm chemical dosing pumps are stroking and day tank levels are consistent with the dosing rate. Record free chlorine and inhibitor residuals from the contractor test kit each day shift, and report out-of-range residuals to the treatment contractor.

6.10 After any power interruption, verify that the duty pump restarted or the standby cut in, and re-establish chemical dosing, which does not restart automatically and is routinely forgotten.

6.11 Blow down the basin to the effluent system when conductivity exceeds the limit on the treatment control sheet, and record the blowdown volume.

6.12 During monsoon operation, inspect the basin screens daily and brief the incoming shift on screen condition at every handover. Wind-blown debris loads rise sharply in the wet season, screen blockage propagates directly to pump suction problems, and in heavy weather the day shift record is stale by night.

6.13 Report any pump vibration, unusual noise or seal weep observed during rounds as a work request against the pump tag, quoting SOP-110 for the vibration assessment.

7. Acceptance Criteria

The system is considered in normal order when one pump carries duty with the standby available in autostart, header pressure and basin level are inside their bands, chemical residuals are in range, and the HX-202A approach temperature shows no rising trend over the trailing week.

8. Records

Shift readings are recorded in the shift log. Monthly duty rotation, residual test results and blowdown volumes are retained on the utilities log sheet and reviewed at the monthly reliability meeting. Any excursion of header pressure, basin level or chemical residuals outside its control band shall be recorded together with the corrective action taken.